

MohammadHosein Shamsipoor

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RESEARCH INTERESTS

Multimodal representation learning; deep generative and latent-variable models (autoencoders, VAEs, normalizing flows, diffusion); representation compression and intrinsic-dimensionality estimation; self-supervised & contrastive learning (CLIP, DINO, MoCo) for vision-language tasks.

EDUCATION

Shahid Bahonar University of Kerman

Kerman, Iran

B.Sc. in Computer Engineering

Sep 2020 – Sep 2025

- Cumulative GPA: **14.94 / 20**; AI/ML & core-CS coursework average: **~17.0 / 20** (see *Selected Coursework*).
- B.Sc. Thesis: *Data-driven estimation of intrinsic dimensionality for autoencoder architecture design* (Advisor: Dr. Mahdi Eftekhari).

RESEARCH EXPERIENCE

Independent Research — Autoencoder Compression & Intrinsic Dimensionality

Self-directed continuation of prior thesis work

June 2026 – present

- Revisiting and extending earlier work on using unsupervised estimates of a dataset's intrinsic dimensionality to size autoencoder bottlenecks/architectures under a target reconstruction-loss threshold.
- Implementing and comparing CNN-autoencoders, MLP-autoencoders, VAEs, and ResMLP models on MNIST, FashionMNIST, and CIFAR-10 (PyTorch). *Writeup in preparation.*

Research Assistant — Machine Learning Group

Kerman, Iran

Shahid Bahonar University of Kerman | Advisor: Dr. Mahdi Eftekhari

Aug 2024 – Sep 2025

- Investigated unsupervised representation-learning methods — autoencoders, non-negative matrix factorization (NMF), basis-driven analysis — to estimate intrinsic dimensionality and inform compact autoencoder design.
- Designed and implemented enhanced autoencoder architectures in PyTorch for representation learning and dimensionality reduction; this work formed the basis of my B.Sc. thesis and an associated manuscript now being revised independently.

PROJECTS

Multi-Condition Synthetic Driving Dataset for Instance Segmentation

collaborative

CARLA + GroundedSAM annotation pipeline

May - June 2025

- Designed and built the full dataset: scripted all four environmental conditions (day, night, rain, fog) in the CARLA simulator, synchronizing a dual-camera (RGB + instance-segmentation) rig at 20 FPS and converting raw contours into YOLO-format polygons; yielded 3,704 frame pairs covering vehicles, pedestrians, buses, and traffic lights (~10.2K, 1.9K, 2.2K instances respectively).
- Built a GroundedSAM-based auto-labeling pipeline (grounded-language prompts → SAM masks) to refine annotations where the simulator's native ground truth was inadequate, plus position-based train/test splitting (for true spatial generalization) and an Albumentations augmentation stage.
- Delivered the dataset to a collaborator who trained/evaluated YOLOv11n-seg and YOLOv11m-seg and compiled the nano model to a TensorRT (FP16) engine, cutting inference latency $\sim 4.1\times$ (11.87 ms → 2.89 ms; 84 → 346 FPS). github.com/Mound21k/yolov11-carla-segmentation

Multi-View Attention for Steering-Angle Prediction

collaborative

ResMLP backbone + multi-camera attention

July - August 2025

- Proposed the approach and implemented the core model — a fusion of a ResMLP backbone with a multi-view attention mechanism over center/left/right camera views — together with the data-processing and training pipeline.
- Project also included a detailed evaluation suite (directional/angular metrics, attention-entropy and cross-view analysis, bootstrap significance testing), built by my collaborator. github.com/Mound21k/MVAM-Steering-Angle

Image Captioning — Show and Tell (PyTorch): implemented CNN-encoder / LSTM-decoder captioning reproducing *Show and Tell* (Vinyals et al.). <https://github.com/Mound21k/image-captioning>

Transfer-Learning Pipeline: pretrained a CNN on a 27k-image ImageNet subset and fine-tuned on CIFAR-100.

INDUSTRY EXPERIENCE

Machine Learning Engineer, Fusion Food Tech / Artechly Dec 2025 – May 2026

Built an AI voice agent for end-to-end order collection; designed order-extraction, verification, and secure automated-payment pipelines.

NLP & AI Engineer, AARTAA Jun 2025 – Sep 2025

Developed transformer-based NLP services (FastAPI, WebSocket); deployed models to production with OAuth 2.0 authentication.

Machine Learning Engineer, Ava Pardaz Keyhan Kariman Feb 2024 – Jul 2024

Fine-tuned T5 models for NER and keyword extraction; built a RAG chatbot and end-to-end NLP retrieval pipelines.

TEACHING EXPERIENCE

Sole Teaching Assistant — Graduate Deep Learning Course Shahid Bahonar University of Kerman

As an undergraduate, under Dr. Mahdi Eftekhari

Feb 2025 – Sep 2025

- Single-handedly designed the full assignment set for a **master's-level** deep-learning course (modeled on graduate DL curricula): an image-captioning project (*Show and Tell*) and a transfer-learning project (ImageNet-subset pretraining → CIFAR fine-tuning).
- Taught self-supervised and vision-language papers (CLIP, DINO, MoCo) and mentored graduate students on neural-network implementations.

Head TA — Artificial Intelligence (Sep 2023 – Jan 2024): led the TA team and designed a capstone multi-agent project (reflex / model-based / goal-based agents competing in a resource-collection grid-world).

TECHNICAL SKILLS

- **Deep Learning & Generative Models:** autoencoders, VAEs, ResMLP, representation learning, self-supervised & contrastive learning (CLIP, DINO, MoCo), CNNs, RNNs.
- **Computer Vision & Multimodal:** vision-language grounding (Grounding DINO), image captioning, synthetic data generation (CARLA), self-supervised representation learning (CLIP, DINO, MoCo).
- **NLP & LLMs:** transformers, foundation models, fine-tuning (T5), sequence modeling.
- **Programming:** Python (proficient), SQL, Rust (basic).
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Hugging Face, Ultralytics, scikit-learn, NumPy, Pandas.
- **Tools & Infrastructure:** Git/GitHub, Linux, Docker, Jupyter.

SELECTED COURSEWORK

(grades out of 20)

Fundamentals of Computational Intelligence (20) • Basics of Speech & Language Processing (17.58) • Fundamentals of Data Mining (17.75) • Computer Vision Basics (16.75) • Special Topics in AI (16.5) • Machine Learning (16.0) • AI & Expert Systems (16.2) • Engineering Probability & Statistics (16.5) • Design of Algorithms (16.5).

CERTIFICATIONS & LANGUAGES

Certifications: ML Specialization (Coursera); NLP Specialization (Coursera); PyTorch for Deep Learning (ZTM).

Languages: Persian (native); English — TOEFL iBT **100/120** (C1). **References:** available upon request.